

1. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

(a) The process $\{X_t : t \in \mathbb{Z}\}$ is such that

$$X_t = 1 + 2t + aX_{t-1} + Z_t$$

where a is a real-valued, non-zero constant. Suppose that the process $\{Y_t : t \in \mathbb{Z}\}$ is defined:

$$Y_t = \nabla^2 X_t$$

where ∇ is the first difference operator. Show that

$$Y_t = aY_{t-1} + Z_t - 2Z_{t-1} + Z_{t-2}.$$

SOLUTION

$$\begin{aligned} \nabla^2 X_t &= \nabla [\nabla X_t] \\ &= \nabla [(1 + 2t + aX_{t-1} + Z_t) - (1 + 2(t-1) + aX_{t-2} + Z_{t-1})] \\ &= \nabla [2 + a(X_{t-1} - X_{t-2}) + Z_t - Z_{t-1}] \\ &= \nabla [2 + a\nabla X_{t-1} + Z_t - Z_{t-1}] \\ &= a\nabla^2 X_{t-1} + Z_t - Z_{t-1} - (Z_{t-1} - Z_{t-2}) \\ &= a\nabla^2 X_{t-1} + Z_t - 2Z_{t-1} + Z_{t-2}. \end{aligned}$$

Hence

$$Y_t = aY_{t-1} + Z_t - 2Z_{t-1} + Z_{t-2}$$

as required.

- (b) Determine the set of values of a for which the process $\{Y_t : t \in \mathbb{Z}\}$ is (weakly) stationary.

SOLUTION

Write the process using the backward shift operator, B , as follows

$$(1 - aB)Y_t = (1 - 2B + B^2)Z_t.$$

The process is (weakly) stationary if the roots of

$$1 - az = 0$$

lie outside the unit circle.

Since the root is $1/a$ it follows that this lies outside the unit circle if $|a| < 1$.

Hence, the process is (weakly) stationary when $|a| < 1$.

- (c) Assuming that a is such that $\{Y_t : t \in \mathbb{Z}\}$ is (weakly) stationary, write this process as a moving average process of infinite order. You should show your working clearly and write your final answer in its simplest form.

SOLUTION

From (b) we have

$$\begin{aligned}(1 - aB)Y_t &= (1 - 2B + B^2)Z_t \\ \implies Y_t &= \frac{(1 - 2B + B^2)}{1 - aB}Z_t.\end{aligned}$$

Let $f(B) = (1 - aB)^{-1}$ then

$$\begin{aligned}f'(B) &= a(1 - aB)^{-2} \\ f''(B) &= 2a^2(1 - aB)^{-3}\end{aligned}$$

Generally, if $f^{(k)}(B)$ is the k^{th} derivative of $f(B)$ then

$$f^{(k)}(B) = k!a^k(1 - aB)^{-(k+1)}.$$

The Taylor series expansion of $f(B)$ about 0 is

$$\begin{aligned}f(B) &= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} B^k \\ &= \sum_{k=0}^{\infty} a^k B^k.\end{aligned}$$

Then

$$\begin{aligned}Y_t &= (1 - 2B + B^2) \left[\sum_{k=0}^{\infty} a^k B^k \right] Z_t \\ &= \sum_{k=0}^{\infty} a^k B^k Z_t - 2 \sum_{k=0}^{\infty} a^k B^{k+1} Z_t + \sum_{k=0}^{\infty} a^k B^{k+2} Z_t \\ &= Z_t + (a - 2)Z_{t-1} + \sum_{k=2}^{\infty} (a^k - 2a^{k-1} + a^{k-2}) Z_{t-k} \\ &= Z_t + (a - 2)Z_{t-1} + (1 - 2a^{-1} + a^{-2}) \sum_{k=2}^{\infty} a^k Z_{t-k} \\ &= Z_t + (a - 2)Z_{t-1} + \frac{a^2 - 2a + 1}{a^2} \sum_{k=2}^{\infty} a^k Z_{t-k} \\ &= Z_t + (a - 2)Z_{t-1} + \frac{(a - 1)^2}{a^2} \sum_{k=2}^{\infty} a^k Z_{t-k}\end{aligned}$$

- (d) Suppose that $a = \frac{1}{2}$. Using your answer to (c), determine the autocovariance function of $\{Y_t : t \in \mathbb{Z}\}$ in terms of σ_z^2 .

SOLUTION

Consider

$$\begin{aligned}\gamma(h) &= \text{Cov}(Y_t, Y_{t+h}) \quad (\text{where } h > 0) \\ &= \mathbb{E}(Y_t Y_{t+h}) - \mathbb{E}(Y_t) \mathbb{E}(Y_{t+h})\end{aligned}$$

Noting that $Y_t = \sum_{j=0}^{\infty} b_j Z_{t-j}$ (from (c)), we see that $\mathbb{E}(Y_t) = 0$ for all t .

Then

$$\begin{aligned}\gamma(h) &= \mathbb{E}(Y_t Y_{t+h}) \\ &= \mathbb{E} \left[\left(\sum_{j=0}^{\infty} b_j Z_{t-j} \right) \left(\sum_{k=0}^{\infty} b_k Z_{t+h-k} \right) \right] \\ &= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} b_j b_k \mathbb{E}(Z_{t-j} Z_{t+h-k})\end{aligned}$$

Since $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process, $\mathbb{E}(Z_{t-j} Z_{t+h-k}) = 0$ except when

$$\begin{aligned}t - j &= t + h - k \\ \implies k &= j + h.\end{aligned}$$

Hence

$$\begin{aligned}\gamma(h) &= \sum_{j=0}^{\infty} b_j b_{j+h} \mathbb{E}(Z_{t-j}^2) \\ &= \sigma_z^2 \sum_{j=0}^{\infty} b_j b_{j+h}.\end{aligned}$$

We note that

$$\begin{aligned}b_0 &= 1 \\ b_1 &= a - 2 = -\frac{3}{2} \\ b_j &= \frac{(a-1)^2}{a^2} a^j = \left(\frac{1}{2}\right)^j \quad (j \geq 2).\end{aligned}$$

For $h = 0$:

$$\begin{aligned}\gamma(0) &= \sigma_z^2 \sum_{j=0}^{\infty} b_j^2 \\ &= \sigma_z^2 \left(1 + \left(-\frac{3}{2}\right)^2 + \sum_{j=2}^{\infty} \left(\frac{1}{2}\right)^j \left(\frac{1}{2}\right)^j \right) \\ &= \sigma_z^2 \left(\frac{13}{4} + \sum_{j=2}^{\infty} \left(\frac{1}{4}\right)^j \right)\end{aligned}$$

$$\begin{aligned}
&= \sigma_z^2 \left(\frac{13}{4} + \frac{1}{16} \frac{1}{(1 - \frac{1}{4})} \right) \\
&= \sigma_z^2 \left(\frac{13}{4} + \frac{1}{12} \right) \\
&= \frac{10\sigma_z^2}{3}.
\end{aligned}$$

For $h = 1$:

$$\begin{aligned}
\gamma(1) &= \sigma_z^2 \sum_{j=0}^{\infty} b_j b_{j+1} \\
&= \sigma_z^2 \left(1 \left(-\frac{3}{2} \right) - \frac{3}{2} \left(\frac{1}{2} \right)^2 + \sum_{j=2}^{\infty} \left(\frac{1}{2} \right)^j \left(\frac{1}{2} \right)^{j+1} \right) \\
&= \sigma_z^2 \left(-\frac{15}{8} + \frac{1}{2} \sum_{j=2}^{\infty} \left(\frac{1}{4} \right)^j \right) \\
&= \sigma_z^2 \left(-\frac{15}{8} + \frac{1}{24} \right) \\
&= -\frac{11\sigma_z^2}{6}.
\end{aligned}$$

For $h \geq 2$:

$$\begin{aligned}
\gamma(h) &= \sigma_z^2 \sum_{j=0}^{\infty} b_j b_{j+h} \\
&= \sigma_z^2 \left(\left(\frac{1}{2} \right)^h - \frac{3}{2} \left(\frac{1}{2} \right)^{h+1} + \sum_{j=2}^{\infty} \left(\frac{1}{2} \right)^j \left(\frac{1}{2} \right)^{j+h} \right) \\
&= \sigma_z^2 \left(\frac{1}{4} \left(\frac{1}{2} \right)^h + \left(\frac{1}{2} \right)^h \sum_{j=2}^{\infty} \left(\frac{1}{4} \right)^j \right) \\
&= \sigma_z^2 \left(\frac{1}{4} \left(\frac{1}{2} \right)^h + \frac{1}{12} \left(\frac{1}{2} \right)^h \right) \\
&= \sigma_z^2 \left(\frac{1}{3} \left(\frac{1}{2} \right)^h \right).
\end{aligned}$$

Since the process is (weakly) stationary, we know that $\gamma(h) = \gamma(-h)$.

Hence

$$\gamma(h) = \begin{cases} \frac{10\sigma_z^2}{3} & h = 0; \\ -\frac{11\sigma_z^2}{6} & |h| = 1; \\ \frac{\sigma_z^2}{3} \left(\frac{1}{2} \right)^{|h|} & |h| \geq 2. \end{cases}$$

2. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

The (weakly) stationary process $\{X_t : t \in \mathbb{Z}\}$ measures the number of celebration cake orders made at a bakery per year, where

$$X_t = 150 + 0.4X_{t-1} + Z_t - 0.5Z_{t-1}.$$

- (a) Show that the mean of this process is 250.

SOLUTION

Taking expectations of random variables we see that

$$\begin{aligned} \mathbb{E}(X_t) &= 150 + 0.4\mathbb{E}(X_{t-1}) + \mathbb{E}(Z_t) - 0.5\mathbb{E}(Z_{t-1}) \\ \implies \mathbb{E}(X_t) - 0.4\mathbb{E}(X_{t-1}) &= 150 \quad (\text{since } \mathbb{E}(Z_t) = 0 \text{ for all } t). \end{aligned}$$

Because $\{X_t\}$ is a (weakly) stationary process, it has a constant mean and it follows that $\mathbb{E}(X_t) = \mathbb{E}(X_{t-1})$. Hence

$$\begin{aligned} (1 - 0.4)\mathbb{E}(X_t) &= 150 \\ \implies \mathbb{E}(X_t) &= \frac{150}{0.6} = 250. \end{aligned}$$

- (b) Show that the process $\{X_t : t \in \mathbb{Z}\}$ is invertible and, hence, write this process in the form

$$Z_t = \sum_{j=0}^{\infty} b_j (X_{t-j} - a)$$

where the coefficients a and b_j ($j \in \{0, 1, 2, \dots\}$) are real-valued constants.

SOLUTION

We define $Y_t = X_t - 250$ and write the process as

$$\begin{aligned} Y_t &= 0.4Y_{t-1} + Z_t - 0.5Z_{t-1} \\ \implies (1 - 0.4B)Y_t &= (1 - 0.5B)Z_t \end{aligned}$$

The process $\{Y_t : t \in \mathbb{Z}\}$ is invertible if the roots of

$$1 - 0.5z = 0$$

lie outside the unit circle. Clearly, this equation has one root at $z = 2$ (which is >1) and, hence, the process $\{Y_t : t \in \mathbb{Z}\}$ is invertible. It follows that $\{X_t : t \in \mathbb{Z}\}$ is invertible.

From the above we write

$$Z_t = (1 - 0.4B)(1 - 0.5B)^{-1}Y_t.$$

If $f(B) = (1 - 0.5B)^{-1}$ then

$$\begin{aligned} f'(B) &= 0.5(1 - 0.5B)^{-2} \\ f''(B) &= 2 \times 0.5^2(1 - 0.5B)^{-3}. \end{aligned}$$

Generally, if $f^{(j)}(B)$ is the j^{th} derivative of $f(B)$ then

$$f^{(j)}(B) = j!0.5^j(1 - 0.5B)^{-(j+1)}.$$

The Taylor series expansion of $f(B)$ about 0 is

$$\begin{aligned} f(B) &= \sum_{j=0}^{\infty} \frac{1}{j!} f^{(j)}(0) B^j \\ &= \sum_{j=0}^{\infty} 0.5^j B^j. \end{aligned}$$

Hence

$$\begin{aligned} Z_t &= (1 - 0.4B) \sum_{j=0}^{\infty} 0.5^j B^j Y_t \\ &= (1 - 0.4B) \sum_{j=0}^{\infty} 0.5^j Y_{t-j} \\ &= Y_t + \sum_{j=1}^{\infty} (0.5 - 0.4) 0.5^{j-1} Y_{t-j} \\ &= Y_t + 0.1 \sum_{j=1}^{\infty} 0.5^{j-1} Y_{t-j}. \end{aligned}$$

Since $Y_t = X_t - 250$ we have

$$\begin{aligned} Z_t &= X_t - 250 + 0.1 \sum_{j=1}^{\infty} 0.5^{j-1} (X_t - 250) \\ &= \sum_{j=0}^{\infty} b_j (X_{t-j} - a) \end{aligned}$$

where

$$\begin{aligned} a &= 250; \\ b_0 &= 1; \\ b_j &= 0.1(0.5^{j-1}) \quad (j \geq 1). \end{aligned}$$

- (c) Observed values of $\{X_t : t \in \mathbb{Z}\}$ for the years 2021–2024 are shown in the table below.

Using this information, produce optimal predictions for the numbers of celebration cake orders for the years 2025 and 2026. Here the term ‘optimal’ implies that the

Year (t)	2021	2022	2023	2024
No. of orders (x_t)	240	268	254	272

predicted values are those that minimise the mean-square prediction error at corresponding time points.

SOLUTION

We know that

$$Y_t - 0.4Y_{t-1} = Z_t - 0.5Z_{t-1}$$

and, supposing that the time series is observed to time n , we define

$$[Z_{n+k}] = \begin{cases} 0 & \text{if } k > 0; \\ Z_t & \text{if } k \leq 0. \end{cases}$$

Then

$$\hat{Y}_{n+k} - 0.4\hat{Y}_{n+k-1} = [Z_{n+k}] - 0.5[Z_{n+k-1}]$$

where $\hat{Y}_t$ is the optimal forecasted value at time t .

Observed values of the series $\{Y_t : t \in \mathbb{Z}\}$ are Consider $k = 1$ (i.e. the year

Year (t)	2021	2022	2023	2024
y_t	-10	18	4	22

2025):

$$\begin{aligned} \hat{Y}_{2025} - 0.4\hat{Y}_{2024} &= [Z_{2025}] - 0.5[Z_{2024}] \\ \implies \hat{Y}_{2025} &= 0.4Y_{2024} - 0.5Z_{2024}. \end{aligned}$$

From (b) we know that

$$Z_{2024} = Y_{2024} + 0.1 \sum_{j=1}^{\infty} 0.5^{j-1} Y_{2024-j}$$

Then, given the observed data,

$$\begin{aligned} Z_{2024} &= Y_{2024} + 0.1Y_{2023} + 0.1(0.5)Y_{2022} + 0.1(0.5^2)Y_{2021} \\ &= 22 + (0.1 \times 4) + (0.05 \times 18) + (0.025 \times -10) \\ &= 23.05. \end{aligned}$$

Finally

$$\begin{aligned} \hat{Y}_{2025} &= (0.4 \times 22) - (0.5 \times 23.05) \\ &= -2.725 \end{aligned}$$

and

$$\hat{X}_{2025} = \hat{Y}_{2025} + 250$$

$$= -2.725 + 250 = 247.275.$$

For $\hat{Y}_{2026}$ we have

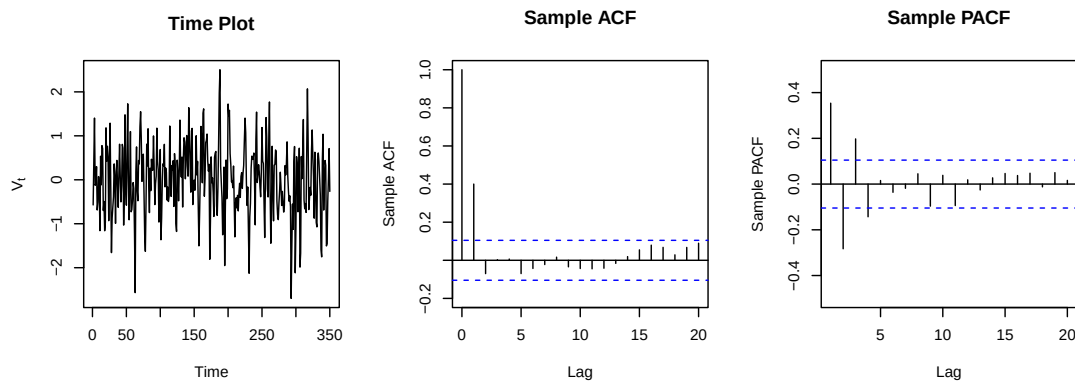
$$\begin{aligned}\hat{Y}_{2026} - 0.4\hat{Y}_{2025} &= [Z_{2026}] - 0.5[Z_{2025}] \\ \implies \hat{Y}_{2026} - 0.4\hat{Y}_{2025} &= 0 \\ \implies \hat{Y}_{2026} &= 0.4\hat{Y}_{2025} \\ &= 0.4 \times -2.725 \\ &= -1.09.\end{aligned}$$

Hence

$$\hat{X}_{2026} = -1.09 + 250 = 248.91.$$

3. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

- (a) A dataset contains 350 sequentially sampled values of the process $\{V_t : t \in \mathbb{Z}\}$. The figure below shows a time plot, a plot of the sample autocorrelation function (ACF) and sample partial autocorrelation function (PACF) against the lag, for this sample.



Using the plots above, suggest an appropriate model for the process $\{V_t : t \in \mathbb{Z}\}$. You should justify your answer and provide approximate estimates of model parameters.

SOLUTION

The time plot indicates that the process is (weakly) stationary, with an apparent constant mean (approximately 0) and constant variance.

The plot of the sample ACF against the lag shows a spike at lag 1 and then values beyond lag 1 are close to zero. This suggests that the process is MA(1) (as the ACF plot 'cuts off' after lag 1).

To confirm this, the PACF gradually declines to zero as the lag increases, which we would expect to see if the process is pure MA.

Hence, an appropriate model for this process is

$$V_t = Z_t + \theta Z_{t-1}.$$

To estimate θ , note that the autocovariance generating function of $\{V_t\}$ is

$$\begin{aligned}\Gamma(z) &= \sigma_z^2(1 + \theta z)(1 + \theta z^{-1}) \\ &= \sigma_z^2(1 + \theta^2) + \sigma_z^2\theta z + \sigma_z^2\theta z^{-1}.\end{aligned}$$

Hence, the ACF at lag 1 is

$$\rho(1) = \frac{\theta}{1 + \theta^2}.$$

Then, with reference to the ACF plot, if we set

$$\frac{\hat{\theta}}{1 + \hat{\theta}^2} = 0.4$$

then

$$\begin{aligned}\hat{\theta} &= 0.4(1 + \hat{\theta}^2) \\ \implies 0.4\hat{\theta}^2 - \hat{\theta} + 0.4 &= 0 \\ \implies \hat{\theta} &= \frac{1 \pm \sqrt{1 - 0.64}}{0.8} \\ \implies \hat{\theta} &= \frac{1 \pm \sqrt{0.36}}{0.8} \\ \implies \hat{\theta} &= 0.5 \text{ or } \hat{\theta} = 2.\end{aligned}$$

For invertibility we choose $\hat{\theta} = 0.5$.

To estimate the variance of Z_t (i.e. σ_z^2), approximately, we see that most of the observed values lie within ± 1 of the mean (0). If we set $2\sqrt{\text{Var}(V_t)} = 1$ then we obtain the estimate

$$\begin{aligned}\hat{\sigma}_z\sqrt{1 + \hat{\theta}^2} &= \frac{1}{2} \\ \implies \hat{\sigma}_z &= \frac{1}{2\sqrt{1 + 0.5^2}} \\ \implies \hat{\sigma}_z &= \frac{1}{\sqrt{5}}.\end{aligned}$$

We conclude that

$$V_t = Z_t + 0.5Z_{t-1}$$

is an appropriate model for $\{V_t\}$.

- (b) A statistician fits the following model to data from a process $\{X_t : t \in \mathbb{Z}\}$:

$$X_t = 0.4X_{t-1} + Z_t - 0.5Z_{t-1} + 0.06Z_{t-2}.$$

The model residuals, $\{Z'_t : t \in \mathbb{Z}\}$, are examined and appear to be an $AR(1)$ process of the form

$$Z'_t = 0.5Z'_{t-1} + W_t,$$

where $\{W_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and a constant variance.

Identify the type of model that the statistician should fit next, including its order. Justify your answer.

SOLUTION

The ‘incorrect’ model is

$$\begin{aligned} X_t &= 0.4X_{t-1} + Z'_t - 0.5Z'_{t-1} + 0.06Z'_{t-2} \\ \implies X_t - 0.4X_{t-1} &= Z'_t - 0.5Z'_{t-1} + 0.06Z'_{t-2} \\ \implies (1 - 0.4B)X_t &= (1 - 0.5B + 0.06B^2)Z'_t \\ \implies X_t &= [\phi_0(B)]^{-1}\theta_0(B)Z'_t. \end{aligned}$$

where

$$\begin{aligned} \phi_0(B) &= 1 - 0.4B; \\ \theta_0 &= 1 - 0.5B + 0.06B^2. \end{aligned}$$

From the question we note that

$$\begin{aligned} Z'_t &= 0.5Z'_t + Z_t \\ \implies (1 - 0.5B)Z'_t &= Z_t \\ \implies Z'_t &= [\phi(B)]^{-1}Z_t \end{aligned}$$

where

$$\phi(B) = 1 - 0.5B.$$

Hence

$$\begin{aligned} X_t &= [\phi_0(B)]^{-1}\theta_0(B)Z'_t \\ &= [\phi_0(B)]^{-1}\theta_0(B)[\phi(B)]^{-1}Z_t \\ &= (1 - 0.4B)^{-1}(1 - 0.5B + 0.06B^2)(1 - 0.5B)^{-1}Z_t. \end{aligned}$$

Therefore

$$\begin{aligned} (1 - 0.4B)(1 - 0.5B)X_t &= (1 - 0.5B + 0.06B^2)Z_t \\ \implies X_t - 0.9X_{t-1} + 0.2X_{t-2} &= Z_t - 0.5Z_t + 0.06Z_{t-2} \end{aligned}$$

and the statistician should fit an $ARMA(2, 2)$ model.

(c) The process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = \frac{1}{4}Y_{t-1} + \frac{1}{8}Y_{t-2} + Z_t,$$

and the process $\{U_t : t \in \mathbb{Z}\}$ is defined

$$U_t = Y_t + \frac{2}{3}Y_{t-1} + \frac{1}{24}Y_{t-2}.$$

Determine the equation for U_t in terms of $\{Z_t : t \in \mathbb{Z}\}$ and past terms of $\{U_t : t \in \mathbb{Z}\}$. What type of process is $\{U_t : t \in \mathbb{Z}\}$?

SOLUTION

We note that

$$\begin{aligned} \left(1 - \frac{B}{4} - \frac{B^2}{8}\right) Y_t &= Z_t \\ \implies Y_t &= \left(1 - \frac{B}{4} - \frac{B^2}{8}\right)^{-1} Z_t. \end{aligned}$$

Then

$$\begin{aligned} U_t &= Y_t + \frac{2}{3}Y_{t-1} + \frac{1}{24}Y_{t-2} \\ &= \left(1 - \frac{B}{4} - \frac{B^2}{8}\right)^{-1} Z_t \\ &\quad + \frac{2}{3} \left(1 - \frac{B}{4} - \frac{B^2}{8}\right)^{-1} Z_{t-1} \\ &\quad + \frac{1}{24} \left(1 - \frac{B}{4} - \frac{B^2}{8}\right)^{-1} Z_{t-2}. \end{aligned}$$

Hence

$$\begin{aligned} \left(1 - \frac{B}{4} - \frac{B^2}{8}\right) U_t &= Z_t + \frac{2}{3}Z_{t-1} + \frac{1}{24}Z_{t-2} \\ \implies U_t - \frac{1}{4}U_{t-1} - \frac{1}{8}U_{t-2} &= Z_t + \frac{2}{3}Z_{t-1} + \frac{1}{24}Z_{t-2} \\ \implies U_t &= \frac{1}{4}U_{t-1} + \frac{1}{8}U_{t-2} + Z_t + \frac{2}{3}Z_{t-1} + \frac{1}{24}Z_{t-2} \end{aligned}$$

and $\{U_t : t \in \mathbb{Z}\}$ is an $ARMA(2, 2)$ process.

4. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

The process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = -\frac{1}{2}X_{t-1} + \frac{3}{16}X_{t-2} + Z_t - \frac{1}{4}Z_{t-1}$$

- (a) Determine the spectral density function of $\{X_t : t \in \mathbb{Z}\}$, expressing your answer in its simplest form as a function of $\cos w$ (where $w \in (-\pi, \pi]$).

SOLUTION

We write

$$\begin{aligned}
 X_t &= -\frac{1}{2}X_{t-1} + \frac{3}{16}X_{t-2} + Z_t - \frac{1}{4}Z_{t-1} \\
 \Rightarrow \left(1 + \frac{B}{2} - \frac{3B^2}{16}\right) X_t &= \left(1 - \frac{B}{4}\right) Z_t \\
 \Rightarrow \left(1 - \frac{B}{4}\right) \left(1 + \frac{3B}{4}\right) X_t &= \left(1 - \frac{B}{4}\right) Z_t \\
 \Rightarrow X_t &= \left(1 + \frac{3B}{4}\right)^{-1} \left(1 - \frac{B}{4}\right)^{-1} \left(1 - \frac{B}{4}\right) Z_t \\
 &= \left(1 + \frac{3B}{4}\right)^{-1} Z_t \\
 &= \alpha(B) Z_t
 \end{aligned}$$

where $\alpha(B) = \left(1 + \frac{3B}{4}\right)^{-1}$

The spectral density function is

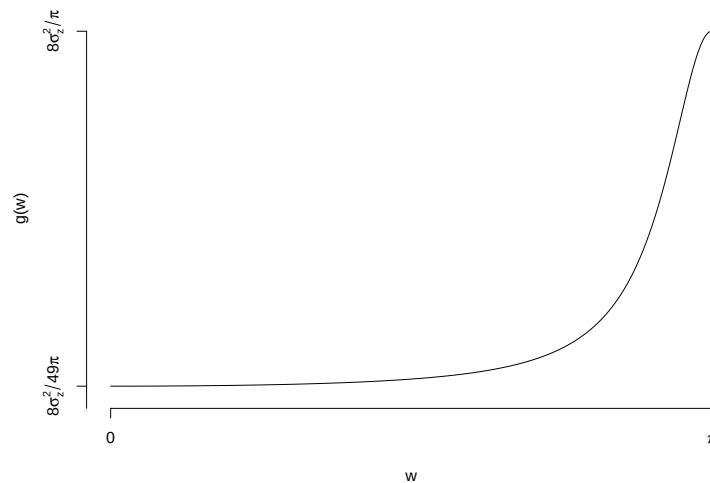
$$g_X(w) = \frac{\sigma_z^2}{2\pi} \alpha(e^{iw}) \alpha(e^{-iw}).$$

Then

$$\begin{aligned}
 g_X(w) &= \frac{\sigma_z^2}{2\pi} \left[\frac{1}{\left(1 + \frac{3e^{iw}}{4}\right) \left(1 + \frac{3e^{-iw}}{4}\right)} \right] \\
 &= \frac{\sigma_z^2}{2\pi} \left[\frac{1}{\left(1 + \frac{9}{16} + \frac{3}{4}(e^{iw} + e^{-iw})\right)} \right] \\
 &= \frac{\sigma_z^2}{2\pi} \left[\frac{1}{\left(\frac{25}{16} + \frac{3}{4}(2 \cos w)\right)} \right] \\
 &= \frac{\sigma_z^2}{2\pi} \left[\frac{1}{\left(\frac{25}{16} + \frac{6 \cos w}{4}\right)} \right] \\
 &= \frac{2\sigma_z^2}{\pi} \left[\frac{1}{\left(\frac{25}{4} + 6 \cos w\right)} \right] \quad w \in (-\pi, \pi].
 \end{aligned}$$

- (b) Sketch the spectral density function from (a) for values of w where $w \in [0, \pi]$. Explain what this sketch suggests regarding the likely behaviour of the process $\{X_t : t \in \mathbb{Z}\}$ over time.

SOLUTION



A sketch of the spectral density function is shown below.

The plot suggests that the series is dominated by high frequency behaviour. This implies that it is not a smooth process and is likely to exhibit rapid fluctuations about the mean, rather than more measured increases/decreases over time.

- (c) A linear filter is applied to $\{X_t : t \in \mathbb{Z}\}$ to create a new process $\{Y_t : t \in \mathbb{Z}\}$, where

$$Y_t = X_t - \frac{9}{16}X_{t-2}$$

Determine the squared gain with respect to this linear filter.

SOLUTION

The transfer function is written

$$h(w) = \sum_{j=-\infty}^{\infty} b_j e^{-i w j}$$

and, here,

$$b_j = \begin{cases} 1 & j = 0; \\ -\frac{9}{16} & j = 2 \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$h(w) = 1 - \frac{9}{16}e^{-i2w}.$$

The squared gain is

$$\begin{aligned} |h(w)|^2 &= \left| 1 - \frac{9}{16}e^{-i2w} \right|^2 \\ &= \left(1 - \frac{9}{16}e^{-i2w} \right) \left(1 - \frac{9}{16}e^{i2w} \right) \end{aligned}$$

$$\begin{aligned}
 &= 1 + \frac{81}{256} - \frac{9}{16} (e^{i2w} + e^{-i2w}) \\
 &= \frac{337}{256} - \frac{18}{16} \cos 2w \\
 &= \frac{337}{256} - \frac{18}{16} (2 \cos^2 w - 1) \\
 &= \frac{625}{256} - \frac{36}{16} \cos^2 w \\
 &= \left(\frac{25}{16} + \frac{3}{2} \cos w \right) \left(\frac{25}{16} - \frac{3}{2} \cos w \right).
 \end{aligned}$$

- (d) Using your answers to (a) and (c), determine the spectral density function of $\{Y_t : t \in \mathbb{Z}\}$.

SOLUTION

The spectral density function of $\{Y_t : t \in \mathbb{Z}\}$ is

$$g_Y(w) = |h(w)|^2 g_X(w).$$

Then

$$\begin{aligned}
 g_Y(w) &= \left(\frac{25}{16} + \frac{3}{2} \cos w \right) \left(\frac{25}{16} - \frac{3}{2} \cos w \right) \times \frac{2\sigma_z^2}{\pi} \left[\frac{1}{\left(\frac{25}{4} + 6 \cos w \right)} \right] \\
 &= \frac{\sigma_z^2}{2\pi} \left(\frac{25}{16} + \frac{3}{2} \cos w \right) \left(\frac{25}{16} - \frac{3}{2} \cos w \right) \left[\frac{1}{\left(\frac{25}{16} + \frac{6}{4} \cos w \right)} \right] \\
 &= \frac{\sigma_z^2}{2\pi} \left(\frac{25}{16} - \frac{3}{2} \cos w \right).
 \end{aligned}$$

- (e) Sketch the spectral density function from (d) and explain briefly how the behaviour of $\{Y_t : t \in \mathbb{Z}\}$ over time differs from that of $\{X_t : t \in \mathbb{Z}\}$.

SOLUTION

A sketch of the spectral density function is shown below. There is a greater density at lower frequencies than was the case for the original process. Hence, we should expect the transformed process to be smoother.

